

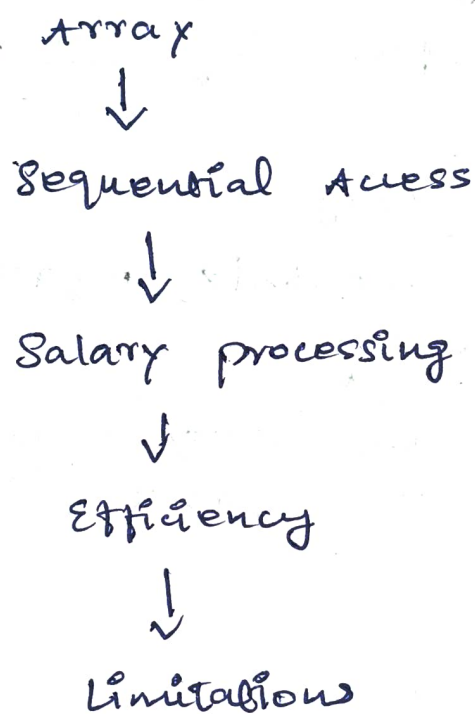
## Concept Mapping

## 27. Payroll System Using Sequential Processing

## Problem statement :-

A company needs to be process the salaries of all employees every month. The employee records are stored in an array.

## Concept Map :-



## Methodology :-

- \* Store Employee data
- \* Sequential Access
- \* Salary processing
- \* Repeat

## Working of Sequential processing

It means the records are accessed in order. The system begins with the first employee in an array & processes every record until the last employee is reached. Every employee is processed exactly once.

## Advantages

- Simple & easy to implement
- Fast when processing all employee records
- Low memory usage
- Suitable for monthly payroll & batch processing
- Accurate because every record is processed in sequence.

## Limitations

- Slow to find a specific employee record as records are checked one by one
- Not suitable for frequent random searches
- Processing time increases as the no. of employees grow.

this method clearly shows how raw data is converted into an efficient software system.

### Solution Process

Smart healthcare System



Data



Data Object



Data Structure



Abstract Data Type



Implementation in C

### Accuracy Calculation

- Structures store patient details together
- Pointers enable dynamic memory management
- ADT operates perform insertion, deletion, searching, updating.

## Interpretation of results

The hierarchy from data → Data Object  
→ Data structure → ADT → Implementation  
in C enables efficient storage, fast  
retrieval, easy updates & better memory  
management of patients records. This  
improves the overall performance,  
reliability & scalability of smart  
healthcare system.

## Result

The payroll system successfully processes employee salaries using sequential access on an array. All employee records are accessed one after another, salaries are calculated correctly, & the payroll is generated efficiently.

## Smart healthcare System

### Problem Understanding

A smart health care system manages large volumes of ~~raw~~ patient data such as temp, blood pressure, medical history & diagnosis. Efficient organization of this data is necessary for quick access, accurate processing & improved health care services.

### Method Selection

A hierarchical concept map is used to represent the relationship between  
Data  $\rightarrow$  Data Object  $\rightarrow$  Data Structure  
 $\rightarrow$  Abstract Data type (ADT)  $\rightarrow$  Implementation  
in C